

OS LAB MANUAL

(CS23431)

Lab:3

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IPC USING SHARED MEMORY

Aim: To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process

Program:

Sender.c:

```
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>
#include <unistd.h>

#define SHM_SIZE 1024

int main() {

    key_t key = ftok("shmfile", 65);
    int shmid = shmget(key, SHM_SIZE, 0666 | IPC_CREAT);
    if (shmid == -1) {

    }

    char *str = (char *)shmat(shmid, NULL, 0);
    if (str == (char *)-1) {
        perror("shmat failed");
    }
}
```



```
shmctl(shmid, IPC_RMID, NULL);  
return 0;  
}
```

Input:

```
Writing to shared memory...  
Message sent successfully!
```

Output:

```
Writing to shared memory...  
Message sent successfully!  
Message Received: hello.what is your name
```